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The effect of signal duration on the underwater detection thresholds of a harbor porpoise (*Phocoena phocoena*) for single frequency-modulated tonal signals between 0.25 and 160 kHz

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The underwater hearing sensitivity of a young male harbor porpoise for tonal signals of various signal durations was quantified by using a behavioral psychophysical technique. The animal was trained to respond only when it detected an acoustic signal. Fifty percent detection thresholds were obtained for tonal signals (15 frequencies between 0.25–160 kHz, durations 0.5–5000 ms depending on the frequency; 134 frequency-duration combinations in total). Detection thresholds were quantified by varying signal amplitude by the 1-up 1-down staircase method. The hearing thresholds increased when the signal duration fell below the time constant of integration. The time constants, derived from an exponential model of integration [Plomp and Bouman, *J. Acoust. Soc. Am.* **31**, 749–758 (1959)], varied from 629 ms at 2 kHz to 39 ms at 64 kHz. The integration times of the porpoises were similar to those of other mammals including humans, even though the porpoise is a marine mammal and a hearing specialist. The results enable more accurate estimations of the distances at which porpoises can detect short-duration environmental tonal signals. The audiogram thresholds presented by Kastelein *et al.* [*J. Acoust. Soc. Am.* **112**, 334–344 (2002)], after correction for the frequency bandwidth of the FM signals, are similar to the results of the present study for signals of 1500 ms duration. Harbor porpoise hearing is more sensitive between 2 and 10 kHz, and less sensitive above 10 kHz, than formerly believed.

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I. INTRODUCTION

The harbor porpoise (*Phocoena phocoena*), one of the smallest cetaceans, has a wide distribution area in the coastal waters of the temperate zone of the northern hemisphere.

The harbor porpoise uses echolocation for detection, ranging and identification of objects (Kastelein *et al.*, 1997, 1999), and for orientation (Verfuß *et al.*, 2005). Its echolocation signals are of short duration [low frequency (LF) clicks 1–3 ms, (Busnel and Dziedziec, 1967; Au *et al.*, 1999); LF clicks: 0.5–5 ms, high frequency (HF) clicks: 90–265 μ s, (Verboom and Kastelein, 2003)]. The distances at which single echolocation clicks, their echoes, and other ecologically important sounds of short duration, such as those made by prey, can be heard by porpoises, can only be assessed with any accuracy if the hearing threshold levels of porpoises for single short signals are known.

Because the harbor porpoise's distribution area overlaps with areas used intensively by humans, it is of interest to

know at what distances sounds from anthropogenic sound sources are audible to porpoises. Some anthropogenic noises are of short duration (*i.e.*, sounds from pile driving, explosions, seismic surveys). To determine whether, and at what distance, short duration anthropogenic noises are audible to harbor porpoises and may disturb them during activities such as foraging, predator avoidance, communication and navigation, information on the harbor porpoise's hearing sensitivity for short duration signals is also needed.

The underwater hearing sensitivity of harbor porpoises has been tested behaviorally for pure tones (Andersen, 1970; 1500 ms including a 50 ms rise and 100 ms fall time), and narrowband FM signals (Kastelein *et al.*, 2002; 2000 ms including 150 ms rise and fall times), and electrophysiologically for clicks, pure tones and noise bursts (Popov *et al.*, 1986; 0.005–100 ms), repeated pure tones (Bibikov, 1992; 5 ms; 10/s), and tone pips and pure tones (Lucke *et al.*, 2007; 0.4–25 ms). Hearing studies on mammals indicate that the ear generally integrates energy over time until the integration time threshold is reached (Watson and Gengel, 1969). Research on the hearing abilities of humans in air (Garner, 1947; Plomp and Bouman, 1959), the harbor seal (*Phoca vitulina*) under water (Terhune, 1988; Kastelein

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et al., 2010), a beluga (*Delphinapterus leucas*) under water (Johnson, 1991), and a bottlenose dolphin (*Tursiops truncatus*) under water (Johnson, 1968) indicate that, in order to be detected, sounds shorter than the integration time of a species must have higher amplitudes than sounds of longer duration of the same frequency.

The aim of the present study was therefore to determine the absolute (unmasked) hearing thresholds of a harbor porpoise for single tonal pulses between 0.25 and 160 kHz and of 0.5–5000 ms duration (depending on the frequency), in order to determine the integration times for those frequencies in porpoises. The results will also indicate optimal signal durations (i.e., >integration time for each test frequency) for future hearing studies with harbor porpoises.

II. MATERIALS AND METHODS

A. Study animal

The male harbor porpoise used in this study (02) was stranded on the Dutch coast (on the island of Texel) at the age of about 21 months and was subsequently rehabilitated. He participated in a previous psychoacoustic study on the detection of tonal signals in noise (Kastelein *et al.*, 2009). During the present study he aged from 3 to 4 years, his body weight varied between 31 and 36 kg, his body length increased from 136 to 142 cm, and his girth in front of the pectoral fins varied between 68 and 75 cm. Veterinary records showed that the animal had not been exposed to ototoxic medication.

The study animal received 2000–3000 g of thawed fish (sprat, *Sprattus sprattus*; herring, *Clupea harengus*; mackerel, *Scomber scombrus*; capelin, *Mallotus villosus* and blue whiting, *Micromesistius poutassou*) per day, equally divided over four meals (three of which were given during research sessions).

B. Study area

The study was conducted at the SEAMARCO Research Institute, The Netherlands, the remote, quiet location of which was specifically selected for acoustic research. Significant noise only reached the study site when, a few times per day, large barges entered a nearby lock, creating LF noise. When this coincided with a hearing test session, the session was stopped, and continued after the barge had left the lock. The study animal was kept in a pool complex built for acoustic research and consisting of an outdoor pool (12 × 8 m; 2 m deep) connected via a channel (4 × 3 m; 1.4 m deep) to an indoor pool (8 × 7 m; 2 m deep; Fig. 1), in which the study was conducted. The pool walls were made of plywood covered with polyester. To reduce sound reflections in the pool (mainly for signals above 25 kHz), the walls were covered with 3 cm thick coconut mats with fibers embedded in 4 mm thick rubber, and the bottom was covered with a 20 cm thick layer of sloping sand on which aquatic vegetation grew. The coconut mats extended to 10 cm above the water level, to reduce splashing sounds produced by waves.

Skimmers kept the water level constant, so that sound conditions were stable. Seawater was pumped directly from

the nearby Oosterschelde, a lagoon of the North Sea, into the open system; recirculation through biological and sand filters ensured year-round water clarity.

The water circulation system and the aeration system for the bio-filter were made as quiet as possible. This was done by choosing “whisper” pumps, mounting the pumps on rubber mats, and connecting the pumps to the circulation pipes with very flexible hoses. The average monthly temperature varied during the year between 0 and 20 °C, and the salinity was around 3.4‰. There was no current in the pool during the experiments, as all pumps were shut off 10 min before and during sessions. By the time a session started, no water flowed over the skimmers, so there was no flow noise during testing.

The equipment used to produce sound stimuli was housed out of sight of the study animal, in a research cabin 4 m away from the underwater listening station. The listening station was at the end of a 3 cm diameter water-filled polyvinylchloride tube. When the study animal was at the listening station, his external auditory meatus was 2 m from the sound source, 1 m below the water surface (Fig. 1).

C. Test stimuli

A schematic of the equipment used to configure and emit outgoing signals is shown in Fig. 2. The porpoise’s hearing sensitivity was tested for tonal signals with center frequencies of 0.25, 0.5, 1, 2, 4, 8, 16, 32, 50, 64, 80, 100, 125, 150, and 160 kHz. Signal durations varied between 0.1 and 5000 ms, depending on the frequency; 134 frequency-duration combinations were tested in all (Table I).

50 000 cycles was the maximum programmable duration the waveform generator (Hewlett Packard, model 33120A) could produce. Therefore, for signals of up to 50 000 cycles, the duration was set by using the “burst” option of the waveform generator (producing pure tones), which allowed the number of cycles produced to be specified. The signals always started and ended with the sinus at zero degrees. To prevent the production of unwanted high frequency onset and offset energy (transients), the tone pulses were filtered by two Butterworth band-pass filters in series [Krohn-Hite, model 3550 (24dB/octave slope); filter set at 1-octave bandwidth around each frequency, and Krohn-Hite, model 3362 (24dB/octave slope), filter set at 2 kHz bandwidth around each frequency].

The minimum signal duration was set at either 0.1 ms, or 5 cycles (depending on the frequency; Table I), because filtering slightly influenced the amplitude of the first and last cycles of the tone burst, and the proportion of cycles in a pulse which was influenced became relatively high below 5 cycles. The longest duration tested was that at which the hearing thresholds for each frequency no longer decreased (i.e., when the integration time was exceeded).

The amplitude of all signals was controlled with an human aerial audiometer (Madsen Electronics, Midimate, model 622, modified to have extended frequency range and signal duration settings). This allowed the sound pressure level (SPL, dB re 1 μPa) at the listening station to be varied in 5 dB increments. The 0.25–4 kHz signals from the audio-

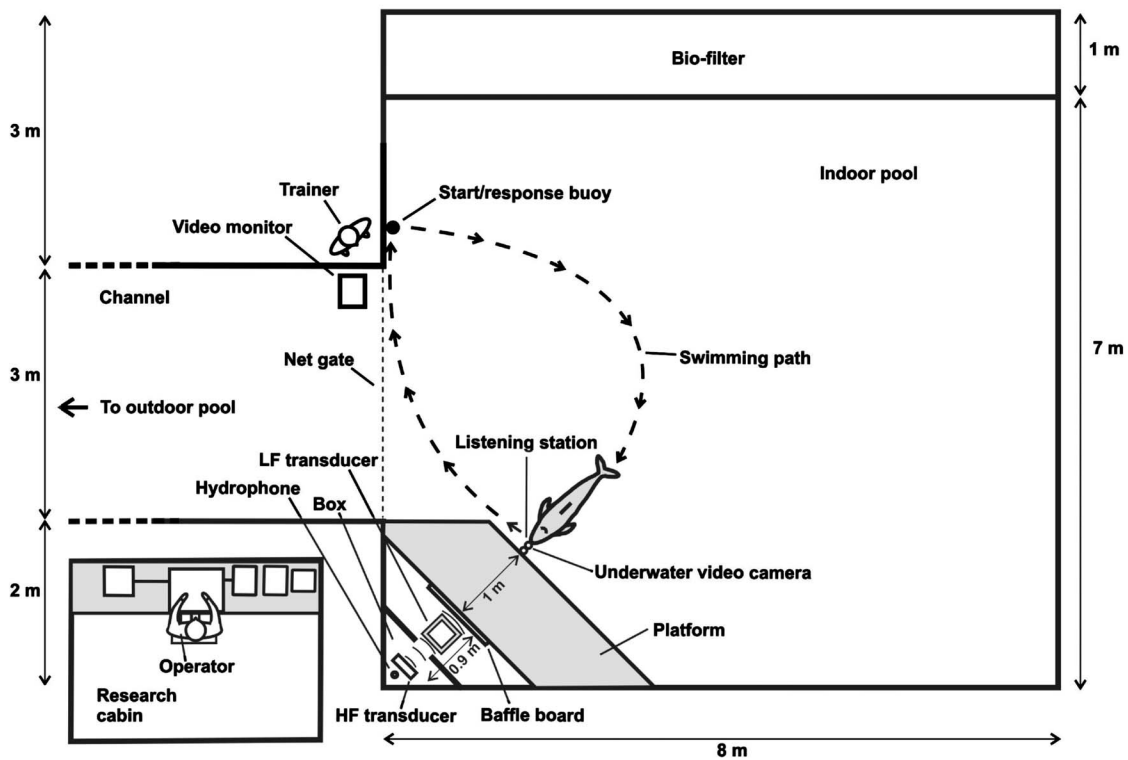
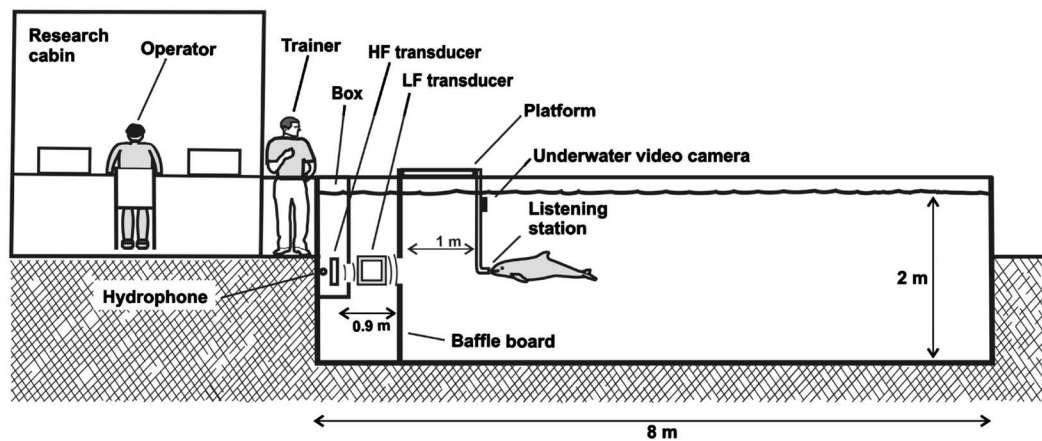
(a) Top view**(b) Side view**

FIG. 1. The study facility, showing the harbor porpoise in position for testing at the listening station, (a) top view and (b) side view, both to scale. Also shown is the swimming path during a trial. The hydrophone shown was used to check the functioning of the sound transmission system, and not for calibration purposes.

meter were also amplified by an audio power amplifier (EHQ, model VPA2450MB, modified with a switch to produce maximum output). Their output was also regulated with two custom-built digitally-controlled attenuators (AS-2008-1, 1 dB steps). Two attenuators were needed to reduce the output power of the amplifier sufficiently for some frequencies.

For signals of 50 kHz and higher, lasting longer than 50 000 cycles, signal durations were controlled by the modified audiometer which had signal duration settings of 1500, 900 and 600 ms including 50 ms rise and fall times (used to prevent abrupt signal onset and offset transients). The steady

state portion of the signal was thus 100 ms less than the reported signal duration. Narrow-band sinusoidal frequency-modulated (FM) tones (warbles) were used for this set up (≥ 50 kHz and > 50 000 cycles). The modulation range of the FM signals was $\pm 1\%$ of the center frequency (the frequency around which the signal fluctuated symmetrically), and the modulation frequency was 100 Hz (for example, if the center frequency was 10 kHz, the frequency fluctuated 100 times per second between 9.9 and 10.1 kHz). Narrow-band FM signals were used because such signals, especially when they are long, are less likely to be dominated by constructive and destructive interference effects (standing

waves) in a reverberant pool than pure tones (Finneran and Schlundt, 2007). 150 and 160 kHz FM signals were filtered (Butterworth, 24 dB/octave slope) so that the bandwidth was 2 kHz around the center frequency. This was done because at

these frequencies on the steep slope of the audiogram, a wide bandwidth could mean that the signal may stimulate the ear at lower frequencies, for which the porpoise is more sensitive.

[illegible]

Two transducers were used to project the signals into the water. The 0.25–4 kHz signals were projected by an inductive moving-coil transducer [NUWC Underwater Sound Reference Division, Newport, RI, U.S.A., model J-11]. The 8–160 kHz signals were projected by a custom-built transducer (WAU, Honolulu, HI, U.S.A., model q7b) consisting of a disk of 1–3 composite piezoelectric materials (Material Systems Inc., Littleton, MA, U.S.A.), with an effective radiating aperture diameter of 4.5 cm. The piezoelectric disk was 0.64 cm thick and 6.4 cm in diameter, and was encapsulated in degassed polyurethane epoxy.

Multi-path sound arrivals can introduce both temporal and spatial variations in the observed SPL at the listening station. To minimize this effect, the WAUq7b transducer was placed in a corner of the pool in a protective wooden box, which was lined with rubber with an irregular surface. The transducer was 1.9 m from the tip of the listening station (Fig. 1), and was positioned so that the acoustic axis of the projected sound beam pointed at the center of the listening station (i.e., at the center of the study animal's head while it was at the listening station). To reduce reflections from the bottom of the pool and the water surface reaching the listening station, a baffle board was placed half way between the transducer and the animal. The board consisted of 2.4 m high, 1.2 m wide, 4 cm thick plywood, covered with a 2 cm thick closed cell rubber mat on the side facing the transducer. A 30 cm diameter hole was made in the board with its center level with the porpoise's head and the transducer (1 m below the water surface). As an indicator of the condition of the transducer, its capacity was checked once a week with a capacity meter (SkyTronic, Nijverdal, Netherlands, model 600.103). Throughout the study period the capacity remained constant.

The exact position of the highly directional small WAUq7b transducer was controlled by attaching it to a plank of 2 cm thick hardwood, which could slide into vertical slits, allowing the transducer to be lowered into the water before each session and removed after each session. The large J-11 transducer did not fit in the sound box, and was therefore placed 1.2 m from the listening station between the box and the baffle board. It was kept exactly in position by being suspended from three cords, just in front of the hole in the baffle board (Fig. 1).

D. Background noise and stimuli level calibration measurements

Great care was taken to make the porpoise's listening environment as quiet as possible. Nobody was allowed to move within 15 m of the pool during sessions. Underwater background noise levels were measured 6 times during the 13-month study period, under the same weather conditions as during the test sessions (no rain, and wind speed Beaufort 4 or below).

The background noise and received levels of the signals were measured using recording and analysis equipment consisting of a B&K 8101 hydrophone with a TPD (custom-built) power supply, a B&K PULSE 3560 D multichannel high frequency analyzer, and a laptop computer with B&K PULSE software Labshop version 12.1. The system was cali-

brated with a B&K 4223 pistonphone. The sound pressure level of each stimulus signal (frequency and duration) was derived from the received 90% energy flux density, divided by the corresponding 90% time duration (Madsen, 2005).

The received SPL (dB re 1 μ Pa) of each frequency-duration combination was measured approximately once each month at the porpoise's head position during the hearing tests (Fig. 1). Each calibration was conducted with two hydrophones, one at each of the locations where the two auditory meata would be when the porpoise was positioned at the listening station during the hearing tests. The background noise was measured at the same locations. The SPL between the two locations varied by 0–3 dB, depending on the frequency. The average received level of the two hydrophones was used for the threshold determinations. During trials, the porpoise's head position (at the listening station) was carefully monitored, and was consistent to within 2 cm for each ear (it was at most 2 degrees off the beam axis of the transducer). The SPL distribution in the $20 \times 20 \times 20$ cm³ area occupied by the head of the porpoise varied by a maximum of 3 dB (depending on the frequency). The received SPLs were calibrated at levels of 40–60 dB (depending on frequency) above the threshold levels found in the present study for the longest signals of each frequency. The linearity of the transmitter system was checked several times during the study; it was consistent to within 1 dB. The test signals used in the study were checked carefully and did not include harmonics at any of the levels at which they were transmitted.

1/3-Octave band sound pressure level spectra of the background noise were determined in the 25 Hz to 160 kHz bands and converted to power spectral density ("spectrum level" expressed in dB re 1 μ Pa²/Hz) by correction for the bandwidth (e.g., Kinsler *et al.*, 1982). The mean ($n=6$) background noise in the pool was very low [Fig. 6(a)]. The background noise was close to the self-noise of the recording equipment.

E. Experimental procedure

A trial began with the study animal at the start/response buoy (Fig. 1). The signal operator had set the frequency and the received SPL for the first trial of the session. The amplitude in the first trial of the session was approximately 15 dB above the detection threshold determined during pre-tests. When the trainer gave a hand signal, the porpoise swam to the listening station (Fig. 1).

To allow the animal's position at the listening station to be checked, he was filmed from above by means of an underwater video camera (Mariscope, model Micro) which was attached to the listening station (Fig. 1). The images were visible to the operator in the research cabin. The porpoise was trained to place the tip of its rostrum at the station and its body axis in line with the beam of the transducer. A maximum deviation of 2° from the beam axis was accepted in all directions. Trials were aborted when the animal was not in the correct position. The trainer indicated this by tapping on

the start/response buoy. When the porpoise returned to the start/response buoy, the trainer sent him straight back to the listening station.

Pre-tests showed that the animal did not need warm-up trials before the actual session began. After the trainer had sent the porpoise to the listening station, he or she moved out of the porpoise's view and watched the porpoise's position at the listening station on a monitor.

In signal-present trials, the porpoise stationed, then had to wait for a period of random duration but between 6 and 12 s (established via a random number generator), before the signal operator produced the test signal. If the animal detected the sound, he was trained to leave the station ("go" response) at any time during the transmission of the signal and return to the start/response buoy [Fig. 1(a)]. When the test signal was produced, a generator was activated that produced horizontal interference lines on the video image. This helped the operator to determine visually whether or not the porpoise responded during production of the signal. If he responded, the signal operator told the trainer that the response was correct, after which the trainer gave the porpoise a fish reward. If the animal did not respond to the sound ("no-go" response) the signal operator would signal this to the trainer. The trainer then signaled to the animal (by tapping three times on the side of the pool) that the trial had ended, thus calling him back to the start/response buoy. No reward was given. If the animal responded before a signal was produced (a prestimulus response), the signal operator would signal to the trainer to ignore the animal for about 10 s. After this short time, a new trial was started by calling the animal to the start/response buoy and then sending it to the listening station. When a prestimulus response was clearly triggered by an external sound which was also detected by the operator, data from the trial were not used. In such cases the trial was repeated as soon as the sound had stopped.

In signal-absent (control or catch) trials, the porpoise stationed, then the signal operator told the trainer after a time period of random duration (but between 6 and 12 s) to end the trial by blowing on a whistle. The animal returned to the start/response buoy and received a fish reward. If the porpoise left the station before the whistle was blown during signal-absent trials, this was regarded as a prestimulus response, and no reward was given. The same amount of fish was given as a reward for correct responses in signal-present and in signal-absent trials. After a trial with a correct response, the next trial would start immediately after the reward had been eaten by the porpoise.

One signal frequency-duration combination was tested in each session. In each session, the signal amplitude was varied according to the 1-up 1-down adaptive staircase method. This conventional psychometric technique (Robinson and Watson, 1973) results in a 50% correct detection threshold (Levitt, 1971). If the animal heard a signal and responded to it (a hit), the signal presented in the next trial was 5 dB lower. If the animal did not hear a signal and remained at the station (a miss), the signal presented in the next trial was 5 dB higher. Prestimulus responses did not result in a change in signal amplitude. A session usually con-

sisted of ~25 trials and lasted for about 15 min. Each session consisted of 2/3 signal-present and 1/3 signal-absent trials offered in random order (determined by using a dice), but without more than three consecutive signal-present or signal-absent trials. In order to end with a positive event, the last trial was always one in which the animal responded correctly and received a reward. This methodology kept the study animal motivated throughout each session (he received a reward after approximately 75% of the trials). Each session, one of four data collection sheets with different random number series was used (to determine the random time between the animal stationing and signal presentation and the random order of signal-present and signal-absent trials). The trainer never knew beforehand whether a trial was a signal-present trial or a signal-absent trial. When the porpoise left the station, the operator observed the animal's behavior on a monitor in the cabin [Fig. 1(a)], told the trainer whether or not to reward the porpoise, and recorded the animal's responses. Other behaviors were solicited from the porpoise between trials to occupy him during occasional short periods of visible or audible disturbances outside the building. Behavioral maintenance sessions, in which no data were collected, were conducted occasionally in order to correct the porpoise's position at the listening station.

Thresholds were determined for 15 test frequencies. Each frequency-duration combination was tested until at least 40 reversal pairs had been obtained per combination (collected over the course of at least four sessions). Before each session, the test duration to be used was randomly selected from the predefined set of durations. The frequencies were tested in the following order (which was influenced by the times at which different pieces of equipment were available; some frequencies and durations were added based on the results during the study): 4, 2, 1, 0.5, 0.25, 32, 64, 80, 100, 125, 150, 160, 50, 8, 16, 4, 2, 1, 0.5, 0.25, 150 and 160 kHz.

Before each session, the voltage output of the emitting system to the transducer and the voltage output of the sound receiving system were checked with an oscilloscope and voltmeter. If they were the same as during the SPL calibration sessions, a test session could begin (Fig. 2). Also, the background noise level was checked to make sure it was not too high for testing (nearly every time they were checked, these conditions were met).

Two to four experimental sessions per day were conducted five days per week (at 0830, 1130, and 1400 h), between January 2009 and February 2010.

F. Analysis

A switch from a test signal amplitude that the porpoise responded to (a hit), to an amplitude that it did not respond to (a miss, indicating that the signal had not been heard), and vice versa, is called a reversal. The mean 50% detection threshold for a frequency-duration combination was determined by calculating the mean amplitude of all reversal pairs in the 4–5 sessions for that test frequency. Sometimes more than the minimum of 40 reversals were collected because all reversals collected in the last session to reach this criterion

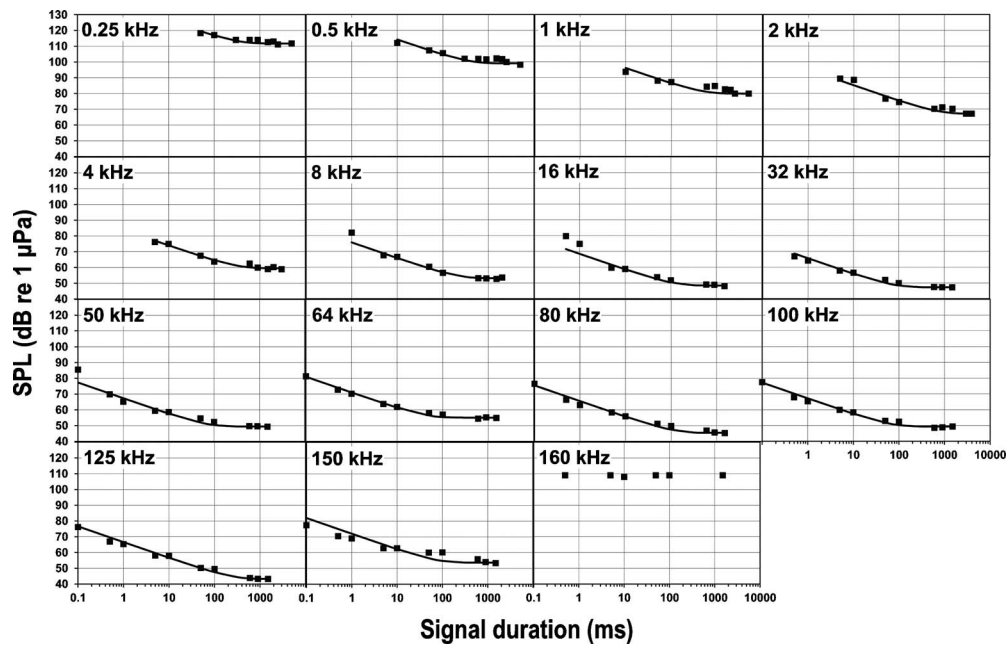


FIG. 3. The relationship between signal duration and mean 50% detection thresholds (dB re 1 μ Pa, rms) in the male harbor porpoise for tonal signals of 0.25, 0.5, 1, 2, 4, 8, 16, 32, 50, 64, 80, 100, 125, 150, and 160 kHz (for details see Table 1). Each data point is based on a minimum of 40 reversal pairs. The curves give the least-squares fits of the mean data to the exponential model (Eq. 6 in Plomp and Bouman, 1959). For each frequency, the time constant of integration (τ) was calculated (see Fig. 5).

were used to calculate the mean threshold. This resulted in a mean number of reversal pairs per test signal frequency-duration combination of 51. Occasional sessions with more than 20% prestimulus responses were eliminated. These sessions were usually simultaneous with external noise caused by distant activities (e.g., aircraft flying or construction), or occurred when the animal was restless. Sessions were only used after the session threshold leveled off, which usually occurred after the first 2–4 sessions with a particular frequency-duration combination. In total 14195 trials were used in the analysis.

All tested received levels for the 1500 ms signals of each test frequency were plotted against the detection rate. From these psychometric functions (of which the sample size per received level varied due to the staircase method of signal presentation), the levels at the 100% detection rates were derived.

To compare the hearing ability of the harbor porpoise with that of other species, the exponential model of temporal

integration (Plomp and Bouman, 1959) was fitted to the mean hearing thresholds for signal frequencies of 0.25 to 150 kHz, using least-squares regression. In logarithmic space, this model can be described by the following equation:

$$10 \log_{10}(I/I_{\infty}) = -10 \log_{10}(1 - e^{-t/\tau}), \quad (1)$$

where τ is the time constant of the porpoise hearing, I is the equivalent plain wave intensity (proportional to the measured mean squared sound pressure) at the detection threshold for a signal with duration t , and I_{∞} is the intensity at the detection threshold of a signal that is much longer than the time constant. Thresholds for very short tones are known to diverge from Eq. (1) because of spectral splatter outside the critical band (Plomp and Bouman, 1959), so some were excluded from the analysis (<0.1 –10 ms, depending on the frequency). The intensity of the longest signal was set as the reference I_{∞} , but if it decreased the sum of squared errors, the mean sound pressure level over the longest two signals was used.

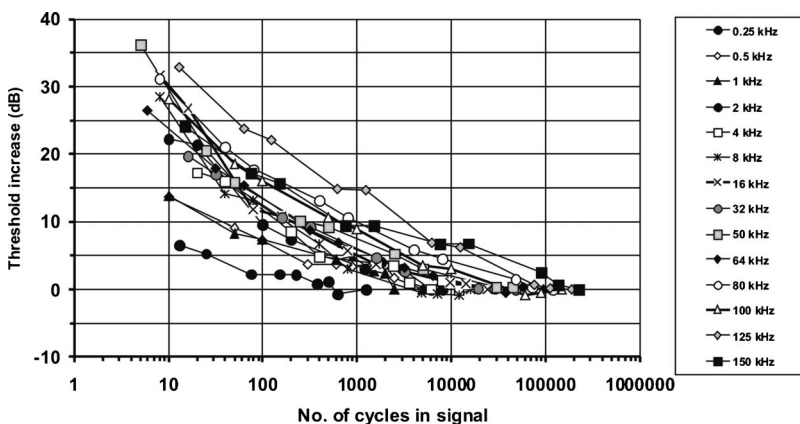


FIG. 4. The relationship between the number of cycles in a pulse and the increase in hearing thresholds in the male harbor porpoise for tonal signals of 0.25, 0.5, 1, 2, 4, 8, 16, 32, 50, 64, 80, 100, 125, and 150 kHz.

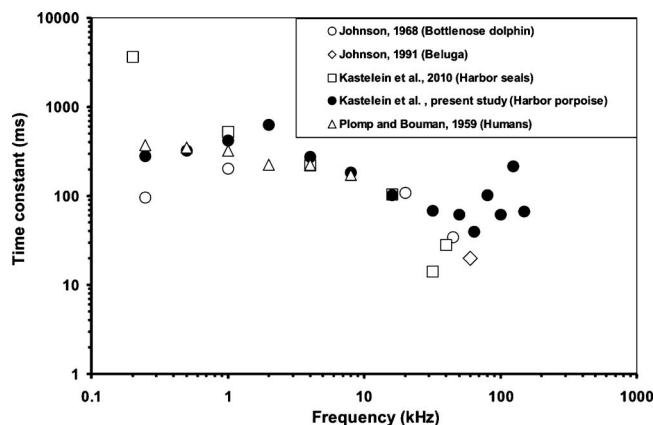


FIG. 5. Integration time constants (τ) based on the mean hearing thresholds of the harbor porpoise in the present study (●), a bottlenose dolphin (Johnson, 1968; ○), a beluga (Johnson, 1991; ◇), two harbor seals (Kastelein *et al.*, 2010; □) and humans (Plomp and Bouman, 1959; △). In general, the higher the frequency, the greater the threshold increases as signal duration decreases.

III. RESULTS

The mean prestimulus response rate (for both signal-present and signal-absent trials) was 6% (SD: $\pm 4\%$, range: 1%–17%). This is considered acceptable for psychophysical hearing studies.

The relationships between signal duration and the 50% hearing threshold level for the harbor porpoise and for the 15 frequencies are shown in Fig. 3 and Table I. The general trend was for thresholds to increase as the pulse durations shorten. For the 160 kHz signals, thresholds remained the same regardless of signal duration.

The relationship between the number of cycles in a pulse and the increase of the hearing threshold levels in the male harbor porpoise are shown in Fig. 4. In general, the higher the frequency, the greater the threshold level increase when signals became shorter.

The integration time constants (τ) based on the mean hearing thresholds of the harbor porpoise are shown in Fig. 5. Fitting the exponential model [Eq. 1 in Plomp and Bou-

man (1959)] to the 50% detection thresholds yielded values for $\tau \pm 95\%$ confidence interval (CI) with r -squared values (Table II). The r -squared values indicate that the model explained the data better for higher frequencies (>4 kHz) than for lower frequencies.

The audiograms based on mean hearing thresholds of the 3–4 year old porpoise for tonal signals of 0.5–5000 ms are U-shaped, with the range of best hearing (defined as 10 dB within maximum sensitivity) from 8 to 150 kHz, and reduced sensitivity around 64 kHz [Fig. 6(b)]. Maximum sensitivity (about 43 dB re 1 μ Pa) occurred at 125 kHz.

The 100% detection thresholds for 1500 ms signals (derived from psychometric functions based on the data from the adaptive staircase method) were on average 6 dB (± 2 dB) above the 50% detection thresholds for the same signal duration.

IV. DISCUSSION

A. Evaluation of the data

The masked hearing thresholds of the study animal in a recent study were similar to those of another porpoise (Kastelein *et al.*, 2009), so the porpoise in the present study probably had normal hearing, and the hearing thresholds found in the present study are probably typical for young harbor porpoises.

For the 160 kHz signals, the detection thresholds were the same for all 6 signal durations tested. This frequency is near the limit of the hearing range and hearing becomes very insensitive over a small frequency increase. At this frequency, the measured hearing threshold may be determined more by the detection of energy below 150 kHz, created during the onset and offset of the signal, than by a change in signal duration.

The high r -squared values (0.93–0.99) and small confidence intervals found for τ indicate that the exponential model fitted the data very accurately at frequencies of 4–100 kHz. It is therefore likely that the decreasing trend in the time constants in this frequency range reflects a physical mechanism of the porpoise ear.

The lower accuracy of the fits at 0.25–2 kHz is mainly caused by a drop in hearing thresholds for longer duration signals (of more than 2 s). Average background noise levels were not high enough to mask the signal [Fig. 6(a)], but it is possible that modulations of the noise (causing peaks in the temporal envelope) may have masked shorter duration signals (Branstetter and Finneran, 2008).

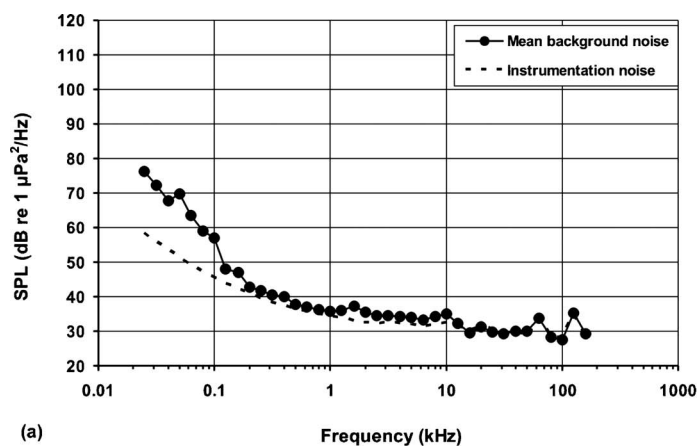
The results of the present study indicate optimal signal durations (i.e., $>$ integration time for each test frequency) for future hearing studies with harbor porpoises.

B. Comparison with previous harbor porpoise hearing threshold study

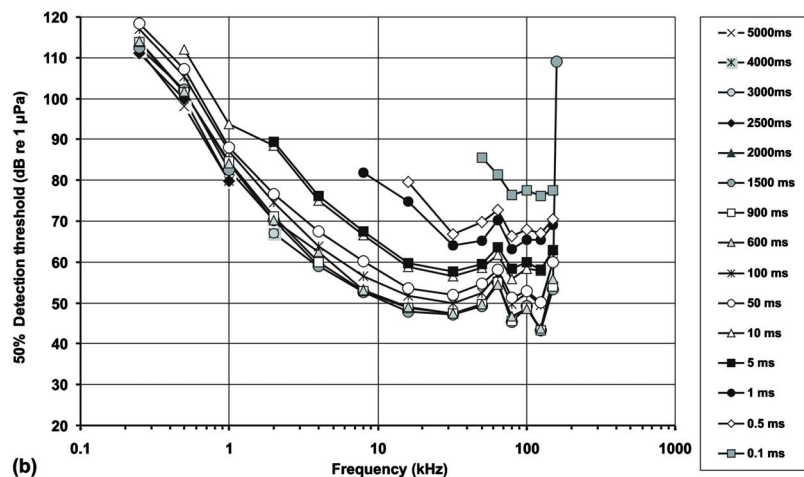
The audiogram of the animal in the present study [Fig. 6(c)] is similar in shape to that of another young male harbor porpoise (Kastelein *et al.*, 2002). Both have maximum sensitivity around 125 kHz, as expected because the echolocation signals of porpoises fall into a narrow band between 120 and 140 kHz (Busnel and Dziedzic, 1967; Møhl and Ander-

TABLE II. The integration time (τ) of the harbor porpoise and its 95% confidence interval, and r^2 of the curve fits.

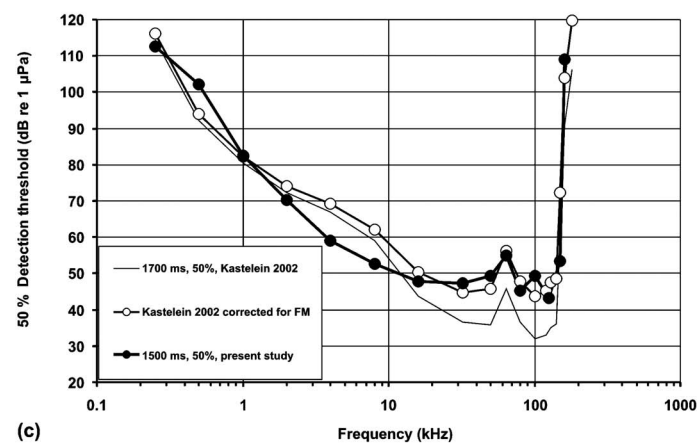
Center frequency (kHz)	τ (ms)	95% range/interval	r^2
0.25	279	± 129	0.74
0.5	320	± 183	0.77
1	417	± 320	0.69
2	629	± 412	0.74
4	277	± 85	0.97
8	185	± 40	0.99
16	102	± 38	0.95
32	68	± 22	0.97
50	62	± 29	0.93
64	39	± 12	0.97
80	102	± 43	0.95
100	61	± 23	0.96
125	215	± 64	0.98
150	67	± 58	0.69



(a)



(b)



(c)

FIG. 6. (Color online) (a) Mean background noise level in the pool between 25 and 160 kHz. The lower dashed line is the electronic noise level of the recording equipment. (b) The mean hearing thresholds of the 3–4 year old porpoise in the present study for tonal signals of 0.5–5000 ms. (c) The 50% detection thresholds of the porpoise of the present study for tonal signals of 1500 ms, the 50% hearing threshold of another 2–3 year old male porpoise for 1700 ms signals as reported by Kastelein *et al.* (2002), and the thresholds of the latter study corrected for the FM stimulus signals.

sen, 1973; Au *et al.*, 1999; Verboom and Kastelein, 2003; Hansen *et al.*, 2008). The decreased hearing sensitivity at 63 kHz found in the previous hearing study (Kastelein *et al.*, 2002) was also found in the present study, which strongly suggests that this is characteristic of harbor porpoises; it may be due to resonance in the outer, middle and inner ear. The animal in the present study also showed a decreased sensitivity at 100 kHz, which was not found by Kastelein *et al.* (2002).

The 50% detection thresholds found for the 1500 ms duration tonal signals in the present study were generally higher than those found for 1700 ms (plus 150 ms on and off ramps) narrow-band FM tonal signals used in the previous study with a young male harbor porpoise (Kastelein *et al.*,

2002). The difference between the two audiograms increases as frequency increases [Fig. 6(c)]. This difference was found to be due to an error in the signal processing of the test signals by Kastelein *et al.* (2002). In the earlier study only FM signals were used. The frequency modulation range of each signal was $\pm 1\%$ of the center frequency (the frequency around which the signal fluctuated symmetrically), and the modulation frequency was 100 Hz. According to the procedure described by Kastelein *et al.* (2002), the signals were digitized at a sample rate of 512 kHz in blocks of 0.2 s and fast Fourier transformed (FFT) into the frequency domain using a Hanning window. The resulting SPL spectrum has a frequency resolution of 5 Hz. The highest peak in the spectrum was selected to determine the SPL and five consecutive

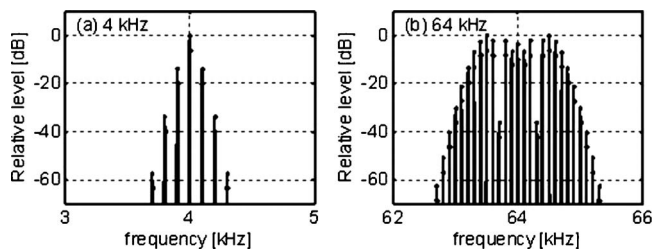


FIG. 7. A synthesized spectrum of a 4 kHz FM signal (a) and a 64 kHz FM signal (b). Note the wide frequency band of the 64 kHz signal due to the proportional (1%) modulation range.

0.2 s time blocks were used to calculate the average SPL. However, selecting the highest peak in the spectrum does not allow the total acoustic energy received at the listening position to be taken into account in the analysis.

Unfortunately, the original measurement data are no longer available. To estimate the error, the signals and analysis procedure were recreated, using numerically simulated FM signals. For each center frequency the FM signal was reconstructed in Matlab™, based on the equation

$$y(t) = \sin\left(2\pi f_c t + 2\pi f_\Delta \int_0^t \sin(2\pi f_m t') dt'\right), \quad (2)$$

in which f_c is the center frequency in Hz, f_Δ the (1%) frequency deviation ($f_\Delta = \frac{1}{100}f_c$) and f_m the modulation frequency (100 Hz).

The FFT of this FM signal exhibits peaks at the center frequency f_c and at side band frequencies $f_c \pm n f_m$, where n represents all positive integers. Peaks outside the FM modulation range $f_c \pm f_\Delta$ are strongly attenuated. Because of the fixed modulation frequency and the proportional frequency deviation, the side bands contribute significantly to the total sound pressure of the signal at center frequencies above 10 kHz (where $f_\Delta > f_m$), while they are attenuated at lower center frequencies. This is illustrated by the spectra of the simulated signals at 4 and 64 kHz center frequency (Fig. 7).

Kastelein *et al.* (2002) selected the highest peak in the power spectrum ($df=5$ Hz) to determine the SPL, but the total received SPL of the signal is actually the sum of the energies in all the peaks. The difference between the two values is frequency dependent. In the simulations, the error (the difference between the single peak level and the total sum) varies between 2 and 13 dB (Fig. 8). This is the theoretical correction for an undistorted signal. In the experiments, the received signals also contained contributions from reflections in the pool, hence the actual difference may be somewhat smaller.

The hearing thresholds of current study animal for 1500 ms tonal signals, of the other 2–3 year old male porpoise for 1700 ms tonal signals (Kastelein *et al.*, 2002), and the same values corrected by means of the method described above, are shown in Fig. 6(c). After correction of the thresholds of Kastelein *et al.* (2002), the audiograms are similar. Apart from individual differences in hearing sensitivity, the difference in thresholds for 160 kHz signals between the two studies occurs, at least partly, because Kastelein *et al.* (2002) used unfiltered FM signals, which produced sound between

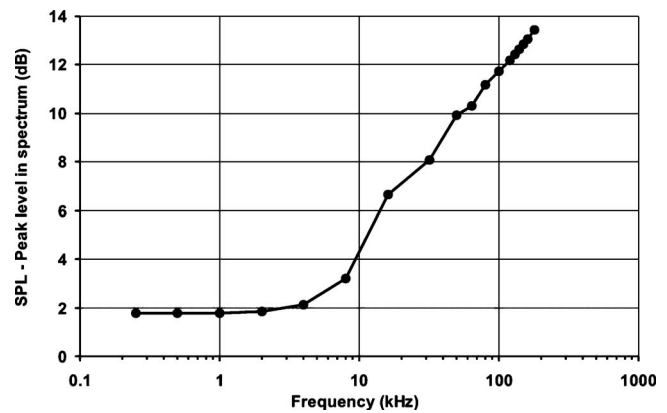


FIG. 8. The difference between the SPL determined by using the highest peak in the spectrum ($df=5$ Hz) of an FM signal, and the actual total received SPL of the signal, determined from the sum of the energies in all the peaks (see Fig. 7).

158.4 and 161.6 kHz, and thus triggered a more sensitive lower-frequency part of the ear. It appears that between 2 and 10 kHz, the hearing thresholds of the porpoise used by Kastelein *et al.* (2002) were masked. It is also possible, though unlikely based on the shape of the curves, that the difference is due to differences in hearing sensitivity between the two porpoises. Thus, based on the thresholds of the present study it can be concluded that harbor porpoises hear better between 2 and 10 kHz, and worse between 10 and 160 kHz, than previously believed.

C. Integration times of other mammals

The integration time constants (τ) of the porpoise in the present study are compared to those of humans (Plomp and Bouman, 1959) and three marine mammal species: the bottlenose dolphin (*Tursiops truncatus*; Johnson, 1968), beluga (*Delphinapterus leucas*; Johnson, 1991), and harbor seal (*Phoca vitulina*; Kastelein *et al.*, 2010), in Fig. 5. Only time constants for single tone bursts are shown. Bullock and Ridgway (1972) found that different brain processes are involved in the processing of broadband clicks and long-duration pure tones. In dolphins, time constants for clicks or periodic tones probably coevolved with the ability to echolocate, and so are much shorter than those for single tones (Au *et al.*, 1988; Johnson, 1991; Au, 1993).

The reported τ values agree quite well; 14 out of 18 data points from earlier studies lie within the 95% confidence intervals of the data from the present study. Only at very low frequencies (≤ 0.25 kHz) and high frequencies (≥ 32 kHz), the harbor seal appears to deviate from the general trend (Fig. 5). The consistency between species suggests that marine mammal integration times for tonal signals are relatively constant, and are comparable to the approximately 200 ms value found for other mammalian species. This suggests that this aspect of mammalian hearing has been conserved in evolutionary terms, despite the different operational environments of, and the various tasks addressed by, the hearing of mammalian species (Gerken *et al.*, 1990).

D. Ecological significance

The results of the present study will enable more accurate estimation of the distances at which porpoises can detect short-duration environmental sounds. The results will also allow the estimation of how loud environmental tonal sounds are perceived to be by harbor porpoises at various distances (received levels usually decrease as distance between the sound source and receiver increases), and under different transmission situations.

Most anthropogenic noises have fluctuating temporal envelopes, and can, partially or completely, mask sounds significant to porpoises such as sounds from predators. Long duration sounds are more likely to be heard within the valleys of the temporal envelope. The results of the present study allow the masking potential of a noise to be determined by calculating the lowest noise levels, occurring over a period of time, which are equivalent to the integration time of the signal frequency of interest.

In the present study, we quantified detection thresholds for tonal signals as a function of signal duration. However, the majority of anthropogenic noise sources are not tonal, and porpoise echolocation signals are far from pure tones. It is not clear how the porpoise hearing integrates complex signals, such as those containing many harmonic components. Further studies, similar to the ones described in this paper, are needed to discover what time constants apply to broadband pulses.

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